Data Landscape — Open Standards for Modern Data Architecture

Curated by Entropy Data · www.data-landscape.com

DEFINITION — HOW DATA IS DESCRIBED

CONTRACTS 6

ADOPT	ADOPT	ADOPT	ADOPT	SITUATIONAL	SITUATIONAL
ODCS BITOL @ LF	OpenAPI LF	AsyncAPI LF	GraphQL LF	gRPC CNCF	OData OASIS

DATA PRODUCTS 4

ADOPT	ASSESS	ASSESS	ASSESS
ODPS BITOL @ LF	DPDS ODM	ODPS LF	DPROD OMG

SCHEMA 7

ADOPT	ADOPT	ADOPT	ADOPT	ADOPT	ASSESS	ASSESS
XML Schema W3C	JSON Schema IETF	SQL DDL ISO/IEC	AVRO Schema ASF	Protobuf GOOGLE	LinkML LINKML	Table Schema FRICTIONLESS

SEMANTICS 9

SITUATIONAL	SITUATIONAL	SITUATIONAL	SITUATIONAL	SITUATIONAL	ASSESS	ASSESS
RDF/OWL W3C	DCAT W3C	SKOS W3C	SHACL W3C	JSON-LD W3C	Ossie Semantic Model OSSIE @ ASF	Ossie Ontology OSSIE @ ASF

ASSESS	ASSESS
schema.org W3C	ShEx W3C

STORAGE — WHERE DATA LIVES

FILE FORMATS 8

ADOPT	ADOPT	ADOPT	ADOPT	ADOPT	ADOPT	SITUATIONAL	ASSESS
CSV IETF	JSON IETF	XML W3C	YAML YAML.ORG	PARQUET ASF	AVRO ASF	ORC ASF	Lance LANCE

OPEN TABLE FORMATS 4

ADOPT	SITUATIONAL	ASSESS	ASSESS
Iceberg ASF	Delta LF	Hudi ASF	Lance LANCE

STORAGE SYSTEMS 2

ADOPT	CAUTION
S3 AWS	HDFS ASF

MOVEMENT — HOW DATA FLOWS BETWEEN SYSTEMS

DATABASE CONNECTIVITY 5

ADOPT	ADOPT	ADOPT	ASSESS	CAUTION
JDBC JCP	ODBC ISO/IEC	ADBC ASF	Quack DUCKDB LABS	XMLA MICROSOFT

INTERCONNECTION 3

ADOPT	SITUATIONAL	CAUTION
HTTP IETF	Delta Sharing LF	FTP / SFTP IETF

MESSAGING 5

ADOPT	ADOPT	SITUATIONAL	SITUATIONAL	CAUTION
Kafka ASF	CloudEvents CNCF	MQTT OASIS	AMQP OASIS	JMS JAKARTA EE

TRANSFORMATION — HOW DATA IS PROCESSED AND RESHAPED

IN-MEMORY FORMAT 2

ADOPT	ADOPT
DataFrame API DATA-APIS	Apache Arrow ASF

PROCESSING 7

ADOPT	ADOPT	ADOPT	ADOPT	SITUATIONAL	ASSESS	CAUTION
Spark ASF	Pandas NUMFOCUS	SQL DML ISO/IEC	dbt DBT LABS	Beam ASF	Ibis IBIS	XSLT W3C

DISCOVERY — HOW DATA IS FOUND AND TRACED

CATALOG APIS 5

ADOPT	ADOPT	SITUATIONAL	ASSESS	CAUTION
Iceberg Catalog ASF	Schema Registry CONFLUENT	Unity Catalog LF	DuckLake DUCKDB LABS	Hive Metastore ASF

LINEAGE 2

ADOPT	ASSESS
OpenLineage LF	PROV W3C

OPERATIONS — HOW DATA IS QUERIED, OBSERVED, GOVERNED

QUERY 5

ADOPT	SITUATIONAL	SITUATIONAL	SITUATIONAL	CAUTION
SQL ISO/IEC	Substrait LF	SPARQL W3C	GQL ISO/IEC	MDX MICROSOFT

DATA QUALITY 3

SITUATIONAL	SITUATIONAL	SITUATIONAL
Great Expectations GX LABS	dbt tests DBT LABS	SodaCL SODA

OBSERVABILITY 2

ADOPT	ASSESS
OpenTelemetry CNCF	OORS BITOL @ LF

POLICIES 3

ADOPT	ASSESS	ASSESS
OPA CNCF	ODRL W3C	OAAS BITOL @ LF

AI INTERFACES 2

ADOPT	SITUATIONAL
MCP LF	A2A LF